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Roll No

EE-7004(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Computer Aided Design of Electrical Machines

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Develop an algorithm for optimization of linear problems. 7
b) Enumerate the general approach for optimization. 7
2. a) Explain the selection of variables for optimal design of D.C. machine. 7
b) Explain the procedure for calculation of main dimensions of D.C. machines. 7
3. a) Distinguish between distribution and Power transformers. 7
b) Explain why Power transformers are designed to have maximum efficiency at or near full load. 7
4. a) How do you design magnetic circuit of a core type Power transformer? Explain it clearly. 7
b) Draw the flow chart for overall design of Power transformer. 7

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5. a) How the variables can be selected for optimal design of an alternator? 7
b) Describe the formulation of design equations of three phase alternator with the help of a suitable examples. 7
6. a) Write an algorithm for determining the overall dimensions of stator of three phase squirrel cage induction motor. 7
b) Calculate diameter and length of stator core and number of turns per phase of a 3-phase 120 kW, 2200 volts, 750 rpm (Synchronous speed) 50 Hz, star connected slip ring induction motor. Give that. 7

$$\text{Average flux density in air gap} = 0.48 \frac{\text{wb}}{\text{m}^2}$$

Specific electrical loading = 26000 ac/m

Efficiency = 92 %

Power factor = 0.88

Winding factor = 0.955

Current density = 5 A/mm²

Pole arc to pole pitch ratio = 1.25

7. a) What do you understand by intuitive solution by computer aided design? Whether human intuition can be programmed in a computer language? Explain its drawbacks. 7
b) Define algorithm. Explain the role of algorithm in computer programming. Explain the different stages of algorithm. 7
8. a) What are the losses in an induction machine? How they can be minimized? 7
b) Explain the phenomenon of logging and crawling in an induction motor. What steps will you suggest at the design to minimize the occurrence of this phenomenon? 7

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